SALBAR: SYSTEM FOR ALLOCATION AND LORA-LINKED BARANGAY ASSESSMENT AND RESPONSE

by

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ABSTRACT

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Not more than one page. You need to summarize everything in one page what your project is all about. The abstract will give an overview of the project. The abstract must be written well in order to arouse the interest of the reader for him/her to continue reading your project. Please discuss this with your adviser.

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ACKNOWLEDGMENTS

This is the section where you will express your gratitude to the people who in one way or another extended their support in your project. You are allowed to use personal pronouns here like I, we, you, etc. You are free to express your feelings here

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CHAPTER 1

INTRODUCTION

1.1 PROBLEM BACKGROUND

The Philippines is consistently recognized as one of the most disaster-prone nations globally, ranking first among 193 United Nations member states in the 2025 World RiskIndex [1]. With an average of twenty tropical cyclones entering the Philippine Area of Responsibility annually, a vast majority of the country's landmass is exposed to intersecting hydro-meteorological and geophysical hazards [2]. Republic Act No. 10121, also known as the Philippine Disaster Risk Reduction and Management Act of 2010, mandates that disaster response be coordinated primarily at the local government level, requiring local government units to maintain dedicated disaster risk reduction and management offices and allocate at least five percent of their estimated revenue for related activities [3]. However, practical implementation remains uneven; nationwide audits indicate that numerous municipalities lack adequate preparation and permanent staff [4]. This institutional disparity is further exacerbated by the country's archipelagic geography, which scatters population centers across varied terrain and complicates both physical access and the establishment of resilient communication infrastructure [5].

The critical problem facing barangay-level disaster response teams and affected residents in Cebu is the severe lack of dynamic, real-time coordination tools during post-disaster operations. Even when baseline communication channels remain intact, the government's formal post-disaster assessment process, such as the Rapid Damage Assessment and Needs Analysis

(RDANA), relies heavily on manual, paper-based exercises completed by trained teams [7]. These traditional methods are oriented toward retrospective, area-level relief planning rather than the immediate, data-driven coordination required to triage emergencies during an active crisis. Because the existing response framework lacks an integrated digital dispatch system, communication and inter-office coordination between barangay- and city-level disaster management units become recurring, documented difficulties [8]. This reliance on fragmented, manual procedures results in delayed, misinterpreted, or entirely unreachable emergency responses, even when rescue personnel and resources are stationed in the immediate vicinity [8].

This fundamental lack of systematic coordination is then severely compounded by an over-reliance on centralized commercial telecommunications grids that frequently fail under extreme conditions [6]. This fragility was starkly illustrated in late 2025 when Cebu was struck by three major disasters within approximately six weeks: a magnitude 6.9 earthquake on September 30 [9], Typhoon Tino, and Super Typhoon Fung-wong [10]. During these events, telecommunications providers experienced outages lasting multiple days across various municipalities, mirroring the widespread disruptions previously recorded during Typhoon Rai (Odette) in December 2021 [11]. When these primary cellular services fail, the already inefficient local response efforts are completely paralyzed. This constraint is most severe in communities formally classified as Geographically Isolated and Disadvantaged Areas (GIDA) [12]. These barangays are separated from mainstream services by difficult terrain or distance, possessing limited baseline connectivity even under normal conditions, and lacking the redundant infrastructure required to maintain emergency operations during commercial outages [13].

To address this critical coordination gap, the System for Allocation and LoRa-Linked Barangay Assessment and Response (SALBAR) provides a comprehensive, infrastructure-independent solution aligned with the research domain of Disaster Resilience and Risk Reduction. To modernize the flawed, manual assessment process, SALBAR introduces a coordination backend that employs a fine-tuned multilingual artificial intelligence model to classify and interpret incident reports in English, Tagalog, and Cebuano, including code-switched text. The system actively resolves logistical bottlenecks by utilizing Google OR-Tools to compute responder allocations that account for responder fatigue and uncertain road conditions, governed by a propose-only AI agent requiring human approval before any dispatch. To ensure these coordination tools remain functional during commercial network outages, SALBAR replaces reliance on fragile cellular grids with a LoRa mesh network that relays messages device-to-device between handheld response-team terminals and a central gateway unit. The field terminals are built on the ESP32-based TTGO T-Beam platform and include an onboard GPS signal filtering and smoothing pipeline. Furthermore, residents can report emergencies through an Android application that sends data directly to the backend when internet connectivity is available; during outages, the app stores and relays reports over Bluetooth to a nearby response-team terminal for forwarding through the LoRa mesh. Through this integration of algorithmic dispatching and infrastructure-independent hardware, SALBAR bridges the gap between isolated communities and centralized rescue operations.

1.2 PROJECT GOAL AND OBJECTIVES

Project Goal

The primary goal of this project is to design and develop SALBAR (System for Allocation and LoRa-Linked Barangay Assessment and Response), an integrated hardware, firmware, and software disaster response coordination system that sustains emergency reporting and responder coordination during cellular and internet disruptions, using an infrastructure-independent LoRa mesh network, a multilingual AI layer, and an operations-research backend.

Objectives

To achieve the goal of the study, the following specific objectives must be met:

1. Design and implement a LoRa mesh communication system linking handheld response-team terminals to a central gateway, including firmware for an OLED and four-button interface for structured triage reporting, GPS-based location tagging with signal filtering and smoothing, and a Bluetooth interface for receiving resident reports.
2. Develop a resident-facing Android application featuring quick-tap triage categories and text input that submits incident reports to the coordination backend when internet connectivity is available, and otherwise stores them on the device and relays them over Bluetooth to a nearby response-team terminal for forwarding through the LoRa mesh.
3. Develop a coordination backend and operator dashboard that classify multilingual disaster reports using a fine-tuned transformer model, generate human-reviewed situation briefings through a constrained AI agent, compute responder allocations using Google OR-Tools under road-condition uncertainty and responder fatigue, and transmit approved dispatch instructions back to the field terminals.

4. Evaluate the system's mesh communication reliability (packet delivery rate, latency, and range), GPS accuracy, multilingual classification accuracy (including code-switched text), responder allocation performance against a first-come, first-served baseline, and overall usability.

1.3 PROJECT SCOPE AND LIMITATIONS

Scope

This study focuses on the design, development, and evaluation of SALBAR, a disaster response coordination system for barangay-level response teams in Cebu. Specifically, the project includes:

- Three handheld response-team terminals built on the ESP32-based TTGO T-Beam platform, assigned to search-and-rescue, damage-assessment, and relief roles, each with filtered and smoothed GPS tagging, an OLED display and four-button interface for structured triage reporting and receiving dispatch instructions, and a Bluetooth interface for receiving resident reports.
- One gateway unit that receives reports over the LoRa mesh and forwards them by serial connection to a host computer.
- A LoRa mesh with store-and-forward relay, operating without cellular or internet connectivity.
- A coordination backend on the host computer that classifies reports in English, Tagalog, Cebuano, and code-switched combinations using a fine-tuned transformer model.

- A propose-only AI agent that generates situation briefings, and a Google OR-Tools engine that computes responder allocations under road-condition uncertainty and responder fatigue.
- An operator web dashboard on the host computer for viewing classified reports and responder locations, approving or rejecting AI proposals, and transmitting dispatch commands back to the field terminals.
- A cloud relay service that receives reports sent directly by resident applications while internet is available, from which the host computer retrieves them.
- A resident-facing Android application featuring quick-tap emergency categories and text input, which sends reports to the cloud relay when internet connectivity is available or, when it is not, stores them offline and relays them via Bluetooth to a response-team terminal.
- Evaluation through simulated field stress trials and timed usability drills within a defined test area.

Limitations

Despite its intended contributions, the study is subject to the following limitations:

- The prototype is limited to three terminals and one gateway due to an approximate budget of ₱20,000, and excludes large-scale or citywide deployment.
- Validation uses simulated trials and drills only, not live testing during an actual disaster.
- The classifier supports only English, Tagalog, Cebuano, and code-switched combinations.
- Classifier accuracy depends on the size and quality of the self-assembled training dataset.
- LoRa range and reliability degrade with physical obstructions and terrain.

- GPS fix reliability depends on satellite visibility and may degrade under dense canopy, rubble, or indoors.
- The terminals are battery-powered, so operating time is limited by battery capacity and power-management firmware.
- The classifier, OR-Tools engine, and AI agent cannot run on the microcontrollers and must run on a host computer connected to the gateway.
- The AI agent is propose-only and cannot execute dispatch without human approval.
- The resident application is Android-only, must be installed before a disaster, and does not support iOS.
- Reports reach the backend directly only while the resident's phone has internet connectivity and the cloud relay is reachable. Otherwise they are delayed until the phone is within Bluetooth range of a response-team terminal, whose range is short.
- Retrieving directly submitted reports requires the host computer to have internet access. The LoRa and Bluetooth path does not.

1.4 SIGNIFICANCE OF THE STUDY

The implementation of the SALBAR project is significant for the following key reasons:

- **Enhancement of Operational and Community Resilience:** The system provides a critical communication lifeline for disaster-stricken communities, particularly Geographically Isolated and Disadvantaged Areas (GIDA), by sustaining emergency coordination even when cellular and internet infrastructures fail. It empowers local barangay response teams with an infrastructure-independent LoRa mesh network, allowing them to submit location-tagged reports and receive dispatch guidance in near

real-time. Furthermore, the multilingual artificial intelligence capability lowers the barrier to emergency reporting by processing Tagalog, Cebuano, English, and code-switched inputs, allowing residents to report naturally during high-stress situations without language constraints. Residents gain a reporting channel that uses the internet when available and a Bluetooth relay when it is not.

- **Modernization of Local Disaster Management Protocols:** The project bridges documented gaps in the Philippine disaster risk reduction and management framework by transitioning post-disaster assessments from retrospective, paper-based exercises into real-time, data-driven operations. By integrating operations-research methods via Google OR-Tools, the system actively accounts for real-world deployment variables, such as responder fatigue and uncertain road conditions, to optimize resource allocation. This integration of onboard GPS processing, applied artificial intelligence, and algorithmic dispatching provides local government units with a proactive, scalable framework to replace historically manual and vulnerable coordination efforts.

1.5 THEORETICAL BACKGROUND

1.5.1 Low-Power Wide-Area Networks (LPWAN) and LoRa Modulation

Long Range, or LoRa, is a physical-layer wireless communication technology used in Low-Power Wide-Area Networks (LPWANs). It uses a proprietary form of Chirp Spread Spectrum (CSS) modulation, in which the carrier frequency changes continuously over time to form chirp signals [14]. This technique allows LoRa communication to maintain reliable transmission over relatively long distances while using low power and narrow bandwidth [14]. LoRa is therefore suitable for emergency communication systems in which message size is

small, energy availability is limited, and commercial cellular or internet infrastructure may be unavailable.

The theoretical performance of a LoRa transceiver is primarily affected by bandwidth (BW), coding rate (CR), and spreading factor (SF) [14, 15]. Bandwidth determines the frequency range occupied by the signal, while coding rate determines the amount of error-correction information included in a transmitted packet. The spreading factor determines the number of chips used to represent each transmitted symbol. In typical LoRa systems, the spreading factor may range from 6 to 12. A higher spreading factor increases processing gain and receiver sensitivity, making the signal more resistant to noise and allowing communication over greater distances [15]. However, it also decreases the data rate and increases the time required to transmit a packet. The relationship between bandwidth and spreading factor can be represented by the approximate symbol-rate equation [15]:

$$R_s = \frac{BW}{2^{SF}}$$

where R_s is the symbol rate in symbols per second, BW is the signal bandwidth in hertz, and SF is the spreading factor. Based on this relationship, increasing the spreading factor increases the denominator 2^{SF}, thereby reducing the symbol rate. This explains why higher spreading factors improve communication robustness but result in slower transmission. The approximate bit rate of a LoRa transmission may also be expressed as:

$$R_b = SF\left(\frac{BW}{2^{SF}}\right)CR$$

where Rb is the approximate bit rate, SF is the spreading factor, BW is the bandwidth, and CR represents the coding-rate factor [14, 15]. The equation demonstrates the trade-off

among communication range, error resistance, and transmission speed [14, 15]. A configuration with a high spreading factor and stronger error correction may be more reliable in an obstructed disaster environment, but it will transmit fewer data bits per second and occupy the radio channel for a longer period [14].

Another important parameter is time-on-air, which refers to the total duration required to transmit one LoRa packet [14]. Time-on-air generally increases as the spreading factor or packet size increases [14]. A longer time-on-air improves the opportunity for the receiver to detect and decode the signal, but it also increases latency, energy consumption, and the possibility of collisions with other transmissions [14]. For SALBAR, messages should therefore contain only essential information, such as the report identifier, incident category, priority, timestamp, location, source terminal, and routing information. The received signal quality is also influenced by the distance between devices, antenna orientation, transmission power, terrain, buildings, vegetation, and other obstructions [14]. In disaster-response environments, damaged structures, steep terrain, dense vegetation, and temporary obstacles can weaken or block the radio signal. Although increasing the spreading factor can improve the probability of successful reception, it cannot completely eliminate the effects of severe obstruction or inadequate antenna placement [14]. In the SALBAR system, LoRa is used to transmit structured incident reports, responder locations, status updates, and approved dispatch instructions between handheld response-team terminals and the gateway. The system prioritizes reliability and coverage over high data throughput because emergency reports are relatively small and do not require continuous high-speed communication. This design is consistent with the operational requirements of barangay-level disaster response, where the successful delivery of a short emergency report is more important than transmitting large quantities of data quickly.

1.5.2 Mesh Topology and Store-and-Forward Routing

Standard LPWAN deployments operate on a star-of-stars topology, requiring end nodes to establish a direct line of communication with a central gateway. However, in heavily obstructed disaster zones, this topology often fails due to line-of-sight blockage. LoRa-Mesh networks resolve these coverage challenges by utilizing decentralized, multi-hop routing protocols [16]. Rather than communicating exclusively with the gateway, the handheld terminals act as intermediary relay nodes, passing packets device-to-device across the network grid [16]. To maintain data integrity when node connections are temporarily lost, the mesh incorporates store-and-forward mechanisms. If a terminal cannot find an immediate path to the gateway, it stores the telemetry packet in its local memory and forwards it only once a viable routing path becomes available [16]. This decentralized self-organization ensures that emergency reports are not dropped during temporary network partitions, making the entire system independent of centralized cellular backbones.

1.5.3 GPS Localization Mechanics and Trilateration

Global Positioning System (GPS) and other Global Navigation Satellite Systems (GNSS) determine a receiver's position using the geometric principle of trilateration. By measuring the time of flight (ToF) of radio frequency signals transmitted from at least four orbiting satellites, a GPS receiver computes its exact three-dimensional coordinates and synchronizes its local clock. However, the theoretical accuracy of this trilateration—often capable of reaching centimeter-level precision under ideal open-sky conditions—severely diminishes when deployed in obstructed environments [17]. In urban canyons, dense forest canopies, or collapsed infrastructure zones, GPS signals suffer from reflection, diffraction, and blockage, transforming clear Line-of-Sight (LOS) signals into Non-Line-of-Sight (NLOS) signals [17]. This

phenomenon, known as multipath fading, causes the receiver to calculate artificially extended ToF values, resulting in significant positional drift and coordinate scattering [18]. For disaster response systems operating in such dynamic and obstructed environments, raw GPS telemetry cannot be reliably used without an intermediate signal processing layer to mitigate these environmental errors. In the SALBAR system, resolving this requires an onboard firmware pipeline to correct and smooth these inaccuracies before transmission over the mesh network.

1.5.4 Signal Filtering and Smoothing Algorithms

To extract accurate location telemetry from noisy and fluctuating raw sensor data, embedded systems utilize mathematical smoothing algorithms. While traditional techniques like the Simple Moving Average (SMA) filter can mitigate high-frequency noise by computing the unweighted mean of the most recent coordinate samples, they are heavily constrained by fixed window sizes and introduce significant phase delays during dynamic movement [19]. A superior alternative for real-time telemetry reconstruction in embedded systems is the Kalman Filter, which is well suited to a robust GPS filtering and smoothing pipeline. Operating as a recursive algorithm for linear quadratic estimation, the Kalman filter relies on a continuous "predict-correct" cycle [20]. In the prediction phase, the filter uses a mathematical state-space model to estimate the current position and its associated uncertainty. In the correction phase, the filter incorporates the newest noisy GPS measurement, updating the state estimate by applying the Kalman Gain—a calculated coefficient that optimally weights the certainty of the theoretical prediction against the reliability of the actual measurement [20, 21]. By naturally filtering out Gaussian noise and adapting to dynamic signal fluctuations without requiring vast computational overhead or unbounded historical data samples, the Kalman filter provides a highly robust method for smoothing raw GPS coordinates at the edge [21].

1.5.5 Transformer-Based Architectures and Self-Attention

For the task of multilingual text classification, modern natural language processing (NLP) systems rely heavily on transformer-based architectures. Unlike earlier recurrent neural networks that process text sequentially, transformers rely entirely on an attention mechanism to weigh the importance of different words in relation to each other, regardless of their position in a sequence [22]. The core of this architecture is the self-attention mechanism, which maps each word in an input sequence to three mathematical vectors: a query, a key, and a value [22]. The attention score between any two words is calculated by taking the dot product of the query vector of the first word with the key vector of the second, which is then normalized using a softmax function to create attention weights [22]. By processing all tokens simultaneously rather than sequentially, transformers can capture complex contextual relationships and long-range syntactic dependencies within text, making them highly effective for rapid and accurate text classification [22]. This mechanism serves as the theoretical foundation for classifying the multilingual incident reports submitted to the SALBAR coordination backend.

1.5.6 Multilingual NLP and Code-Switching

Deploying NLP models in the Philippines presents unique computational challenges due to the pervasive linguistic practice of code-switching, which involves alternating between two or more languages within a single utterance [23]. In the context of disaster reporting, users frequently combine English, Tagalog, and Cebuano into unstructured, mixed-language texts. Mainstream NLP systems often struggle with this practice, treating code-switched input as noisy data rather than a structured bilingual phenomenon [23]. The primary computational challenges

include a severe lack of annotated multilingual corpora, high orthographic variation, and morphological blending (e.g., attaching Filipino affixes to English stems) [23]. To mitigate these issues, systems must employ fine-tuned multilingual transformer models that leverage cross-lingual transfer learning and contextual embeddings. By analyzing the contextual sequence rather than relying on static, single-language dictionaries, these methodologies can accurately interpret the semantic intent of mixed-language text in real-time [23].

1.5.7 Human-in-the-Loop (HITL) and Agentic AI

As artificial intelligence transitions from purely predictive models to "Agentic AI"—systems capable of independent planning and execution—the requirement for strict operational governance becomes critical [24]. In disaster coordination systems, relying entirely on autonomous AI to dispatch responders introduces unacceptable risks, as oversight failures have immediate, real-world consequences [24]. To ensure operational safety, systems can integrate a Human-in-the-Loop (HITL) architecture. HITL is an AI governance approach where trained humans retain ultimate decision authority over high-risk agent actions [24]. Unlike "Human-on-the-Loop" (HOTL) models, where an AI acts autonomously while a human monitors outputs after the fact, a strict HITL system requires the human operator to authorize an action before the AI executes it [24]. In the SALBAR system, the AI agent is constrained to a "propose-only" role: it generates concise situation briefings and proposes responder allocations, but execution is halted until human approval is explicitly granted. This framework mitigates the risk of automation complacency and ensures that high-stakes emergency decisions benefit from human judgment [24].

1.5.8 Constraint Programming and Combinatorial Optimization

Constraint Programming (CP) is a declarative mathematical paradigm utilized to solve complex combinatorial optimization problems, particularly in logistics, scheduling, and resource assignment [25]. Unlike imperative programming, where a strict sequence of steps is defined to compute a solution, CP relies on modeling the problem's state space through decision variables, their allowed domains, and a rigid set of constraints that must be universally satisfied [25]. Mathematical optimization engines, such as the open-source Google OR-Tools suite, employ advanced CP solvers and vehicle routing libraries to evaluate these combinatorial spaces [26]. Because an exhaustive evaluation of every possible responder dispatch combination in a disaster scenario results in an exponential time complexity (an NP-hard problem), OR-Tools utilizes heuristics and metaheuristics—such as Guided Local Search or Tabu Search—to rapidly traverse the solution space, escape local minima, and converge on a globally optimal or near-optimal resource assignment [26]. For the SALBAR system, this constraint-based architecture forms the algorithmic core of the coordination backend, computing the most efficient assignment of available response personnel to incoming incident reports.

1.5.9 Dynamic Resource Allocation Under Uncertainty

Standard routing and dispatching algorithms traditionally assume deterministic environments where travel times, road networks, and agent capacities are known and constant. However, post-disaster environments are highly stochastic, rendering traditional deterministic models dangerously ineffective [27]. Dynamic resource allocation addresses this by employing frameworks like the Stochastic Vehicle Routing Problem (SVRP) or robust optimization models [27]. In these theoretical models, environmental unpredictability—such as uncertain road conditions, flooding, or debris blockage—is mathematically quantified by assigning probabilistic weights, variance penalties, or time-dependent variables to the edges of the navigation graph

[28]. Similarly, human factors like responder fatigue are modeled as dynamic capacity constraints, where a responder's efficiency metric decays over time or degrades after completing consecutive high-intensity tasks [28]. By explicitly incorporating these stochastic variables as operational constraints, algorithmic dispatch systems avoid assigning exhausted personnel to critical missions or routing teams through impassable terrain [28]. Within SALBAR, these theoretical models are implemented via the Google OR-Tools engine to dynamically adjust responder allocation recommendations based on real-world fatigue and road-condition uncertainty.

1.5.10 Decentralized Emergency Triage

Traditional DRRM frameworks often rely on centralized, hierarchical reporting structures. In the Philippines, the formal post-disaster assessment process—such as the Rapid Damage Assessment and Needs Analysis (RDANA)—is traditionally a structured, paper-based, and retrospective exercise conducted by trained teams days after the initial impact [29]. While effective for long-term recovery and large-scale relief planning, this method introduces severe latency during the critical 72-hour emergency response window. Decentralized, data-driven coordination models represent a paradigm shift in emergency triage, transitioning from retroactive data collection to real-time, continuous situational awareness [30]. By distributing the reporting capabilities directly to frontline responders and leveraging edge-computing networks, decentralized models eliminate the single points of failure inherent in strict, top-down chains of command [30]. Within the SALBAR architecture, this theoretical shift is operationalized by replacing manual reporting with an infrastructure-independent digital triage system, allowing isolated response teams to share location-tagged reports in near real-time and, once operators approve the proposed allocations, receive coordinated dispatch guidance.

1.5.11 Infrastructure Vulnerability in GIDAs

Geographically Isolated and Disadvantaged Areas (GIDA) represent communities that are separated from mainstream socioeconomic services due to physical distance, challenging terrain, and a lack of reliable transportation or communication infrastructure [31]. The risk profile of a GIDA is characterized by compounding vulnerabilities: not only are these areas heavily exposed to multiple hydro-meteorological and geophysical hazards, but their baseline connectivity is severely limited even under normal operational conditions [31]. When a major disaster disrupts primary telecommunications grids, GIDAs lack the redundant infrastructure necessary to maintain communication, inevitably leading to information blackouts and delayed rescue operations [32]. Traditional commercial disaster alert systems and mobile applications, which rely entirely on functional cellular networks, systematically fail these communities when they are needed most [32]. Consequently, developing infrastructure-independent communication frameworks, such as the LoRa mesh network proposed by SALBAR, is theoretically and practically essential to mitigate the disproportionate disaster impacts experienced by populations in these vulnerable areas.

1.5.12 Offline-First Architecture and Store-and-Forward on Mobile Devices

Conventional mobile reporting applications depend on a live connection to send every submission to a server, so a lost connection means a lost report. An offline-first architecture reverses this assumption: the device keeps the primary copy of the data, saves input immediately, and treats network access as optional rather than required [37]. To move the saved data onward, delay-tolerant networking uses store-and-forward switching, in which nodes hold a message in storage until the next link becomes available [38]. Opportunistic networking

extends this to mobile nodes, which build routes dynamically and use any node they encounter as the next hop toward the destination [39]. In SALBAR, the resident application saves each report locally with a unique identifier, timestamp, and location. If internet is available, it sends the report to the cloud relay. Otherwise it holds the report until the phone is within Bluetooth range of a response-team terminal, which forwards it through the LoRa mesh as described in Section 1.5.2. The identifier lets the backend discard duplicates if a report arrives by both paths.

1.5.13 Bluetooth Low Energy

Bluetooth Low Energy (BLE) is a short-range wireless technology designed for low power consumption, in which performance reflects a trade-off among energy use, latency, and throughput [40]. It suits small, infrequent exchanges rather than continuous high-volume transfer. Data are exchanged through the Generic Attribute Profile (GATT), where a peripheral exposes data that a central device can read or write, and the default attribute payload is only about 20 bytes unless larger sizes are negotiated [41]. The ESP32 on the TTGO T-Beam includes a BLE-capable Bluetooth radio [42], so the terminal can receive phone reports without additional hardware. In SALBAR, the terminal acts as the peripheral and the resident's phone as the central device that writes a compacted report to it, which the terminal then queues for the LoRa mesh. Because BLE range is short, generally tens of meters, a transfer succeeds only while a resident and a response-team member are near each other.

1.6 REVIEW OF RELATED LITERATURE

In the study of Nguyen et al. entitled "A LoRaWAN-Based IoT System for Post-Disaster Environmental Monitoring" [33], the researchers developed a low-cost sensor network designed to transmit structural health data and emergency distress signals after an earthquake. The study

found that LoRa modulation achieved successful packet transmission over distances exceeding 5 kilometers in dense urban environments without relying on commercial cellular networks. However, the architecture utilized a standard star-of-stars topology, meaning all end-nodes had to communicate directly with a central, internet-connected gateway. When the central gateway was simulated to fail during a secondary shock, the entire network collapsed, and data was permanently lost. SALBAR differs significantly from this approach by implementing a decentralized LoRa mesh topology rather than a star network. By equipping handheld terminals with store-and-forward relay capabilities, SALBAR ensures that if a direct line to the gateway is obstructed, messages will hop device-to-device to maintain network integrity, making it drastically more resilient than traditional gateway-dependent LoRa setups.

The challenge of processing disaster-related communications has been widely explored in computational linguistics. In the study of Cruz and Cheng entitled "Evaluating Language Models for Philippine Disaster Twitter Data" [34], the investigators utilized pre-trained natural language processing algorithms to categorize crowdsourced emergency requests during Typhoon Vamco. The study demonstrated that while standard transformer models achieved over 90% accuracy when classifying strictly English text, their performance degraded sharply—falling below 60% accuracy—when presented with code-switched inputs combining Tagalog and English. The researchers concluded that mainstream models treat code-switching as spelling errors, severely limiting their utility in Philippine rescue operations. SALBAR addresses this specific linguistic limitation by employing a fine-tuned transformer model explicitly trained to classify and interpret English, Tagalog, and Cebuano, including heavily code-switched phrasing. Unlike the retrospective social media analysis conducted by Cruz and Cheng, SALBAR embeds this

multilingual AI directly into a real-time coordination backend to actively triage field reports as they arrive.

Furthermore, existing operations-research models for emergency logistics frequently struggle to account for the stochastic nature of disaster zones. In the study of Zheng et al. entitled "Dynamic Ambulance Routing and Allocation in Post-Disaster Environments" [35], the authors utilized integer linear programming to dispatch medical responders to mass-casualty incidents. Their algorithmic results successfully minimized average response times by 22% compared to manual dispatching. However, a major limitation of their model was the assumption of deterministic environmental variables; the algorithm assumed road conditions remained static and that human responders possessed infinite operational stamina. The researchers noted that in real-world deployments, these assumptions lead to routing failures and exhausted personnel. SALBAR builds upon and diverges from this methodology by incorporating real-world constraints directly into its Google OR-Tools allocation engine. Instead of assuming static variables, SALBAR's backend explicitly computes for road-condition uncertainty and responder fatigue, ensuring that the AI agent only proposes dispatch routes that are operationally viable for the human response teams on the ground.

Finally, attempts to integrate mobile reporting systems into local government operations have faced severe infrastructural bottlenecks. In the study of Reyes and Villanueva entitled "E-Bayanihan: A Web-Mobile Framework for Community Disaster Reporting" [36], the investigators developed an Android application allowing residents to submit location-tagged hazard reports to a barangay dashboard. While usability testing showed high user satisfaction and

rapid report submission times, the system's fundamental flaw was its absolute reliance on commercial 4G/LTE networks. During field tests simulating a typhoon-induced blackout, the application was entirely rendered useless. SALBAR directly addresses this vulnerability. Alongside a central coordination dashboard, it provides a resident-facing Android application that sends reports directly to the backend whenever internet connectivity is available. When it is not, the application stores reports on the phone and relays them over Bluetooth to a response-team terminal, which forwards them through the LoRa mesh. This preserves app-based reporting during partial or complete cellular and internet outages.

1.7 DEFINITION OF TERMS

- **Code-switching** – The practice of alternating between two or more languages—specifically English, Tagalog, and Cebuano—within a single submitted emergency report, which the system's artificial intelligence is designed to classify and interpret.
- **Constrained AI Agent** – An artificial intelligence module within the coordination backend that generates situation briefings and proposes responder allocations, but is strictly restricted from executing any dispatch commands without explicit human approval.
- **First-come, First-served (FCFS) Baseline** – The standard, non-optimized method of dispatching responders based strictly on the chronological order of received reports, used in system evaluation as a comparative benchmark against the OR-Tools allocation engine.

- **Gateway Unit** – The central receiving hardware node of the LoRa mesh network that aggregates incoming incident reports from all field terminals and forwards the data to the host computer for backend processing.
- **Geographically Isolated and Disadvantaged Areas (GIDA)** – Communities separated from mainstream services by distance, difficult terrain, or limited transportation and communication access, serving as the primary target environment for the system due to their vulnerable baseline connectivity.
- **LoRa Mesh Network** – A decentralized, infrastructure-independent wireless communication topology where handheld terminals relay messages device-to-device to reach a central gateway, operating entirely without commercial cellular or internet service.
- **Packet Delivery Rate (PDR)** – A key testing metric used to evaluate the communication reliability of the LoRa mesh network, representing the percentage of successfully transmitted messages that successfully reach the gateway.
- **Resident-Facing Mobile Application** – An Android application that allows residents to submit incident reports to the backend when internet connectivity is available or, when it is not, to store them on the device and relay them via Bluetooth to a response-team terminal for forwarding through the LoRa mesh.
- **Responder Allocation Engine** – The algorithmic backend module that assigns available disaster response personnel to specific emergencies, explicitly computing for real-world constraints such as responder fatigue and uncertain road conditions using the Google OR-Tools library.

- **Signal Filtering and Smoothing Pipeline** – A multi-stage firmware process implemented on the handheld terminals that corrects, filters, and averages raw GPS coordinates to improve location tagging accuracy before the data is transmitted over the mesh network.
- **Store-and-Forward Transmission** – A network relay mechanism where a handheld terminal temporarily saves a message if a direct connection is unavailable and transmits it once another node comes into range, ensuring data integrity during connectivity drops.
- **Transformer-Based Model** – The specific underlying architecture of the natural language processing artificial intelligence classifier used in the coordination backend to interpret and categorize multilingual disaster reports.
- **TTGO T-Beam** – The specific ESP32-based hardware development platform utilized as the foundation for the handheld response-team terminals, featuring integrated LoRa transceivers and GPS modules.
- **Bluetooth Relay** – Transfer of stored resident reports over Bluetooth to a nearby response-team terminal, which forwards them through the LoRa mesh to the gateway.
- **Host Computer** – The computer connected to the gateway by serial link that runs the coordination backend and operator dashboard.

REFERENCES

[1] "Philippines tops WorldRiskIndex 2025 as world's most disaster-prone country," Manila Bulletin, Sep. 26, 2025. [Online]. Available: https://mb.com.ph/2025/09/26/philippines-tops-worldriskindex-2025-as-worlds-most-disaster-prone-country

[2] Philippine Atmospheric, Geophysical and Astronomical Services Administration (PAGASA), "Tropical Cyclone Information," DOST-PAGASA, 2025. [Online]. Available: https://bagong.pagasa.dost.gov.ph/climate/tropical-cyclone-information

[3] "Republic Act No. 10121," Official Gazette of the Republic of the Philippines, May 27, 2010. [Online]. Available: https://www.officialgazette.gov.ph/2010/05/27/republic-act-no-10121/

[4] Commission on Audit (COA), "Disaster Risk Reduction and Management (DRRM) at the Local Level," Sectoral Performance Audit Report, 2020. Available: https://wrd.unwomen.org/sites/default/files/2023-02/PHILIPPINES_NDRRMP_2020-2030.pdf

[5] "Internal and External Constraints on Philippine Local Government," ResearchGate, Aug. 31, 2025. [Online]. Available: https://www.researchgate.net/publication/395113086_Internal_and_External_Constraints_on_Philippine_Local_Government

[6] N. D. R. R. M. C., "National Disaster Risk Reduction and Management Plan (NDRRMP) 2020-2030," Department of National Defense, 2020. Available: https://wrd.unwomen.org/sites/default/files/2023-02/PHILIPPINES_NDRRMP_2020-2030.pdf

[7] Office of Civil Defense (OCD), "Rapid Damage Assessment and Needs Analysis (RDANA) Manual," National Disaster Risk Reduction and Management Council. [Online].

[8] I. Galorio, A. P. Cesa, and M. R. Quilang, "Collaborative governance: Multifaceted strategies for disaster response in Cebu province," Journal of Contemporary Research in Social Sciences, vol. 7, no. 2, pp. 156–163, Dec. 18, 2025. [Online]. Available: https://learning-gate.com/index.php/2641-0249/article/view/11505/3641

[9] "Cebu quake: Key facts and updates," INQUIRER.net, Oct. 03, 2025. [Online]. Available: https://newsinfo.inquirer.net/2119332/cebu-quake-key-facts-and-updates

[10] G. Bandi-Phillips, "Storm Fung-wong: Philippines faces second tropical cyclone in as many weeks," World Food Programme, Nov. 8, 2025. [Online]. Available: https://www.wfp.org/stories/storm-fung-wong-philippines-faces-second-tropical-cyclone-many-weeks

[11] E. Marcelo, "Odette leaves millions without power, telco signals," Philstar.com, Dec. 19, 2021. [Online]. Available: https://www.philstar.com/nation/2021/12/19/2148786/odette-leaves-millions-without-power-telco-signals

[12] Department of Health (DOH), "Guidelines on the Identification and Geotagging of Geographically Isolated and Disadvantaged Areas (GIDA)," DOH Administrative Order No. 2020-0023, 2020.

[13] "Is a geographically isolated community prepared—or left behind—in disaster risk communication?," Davao Research Journal. [Online]. Available: https://davaoresearchjournal.ph/index.php/main/article/view/431/213

[14] H. A. Maleki et al., "A Tutorial on Chirp Spread Spectrum Modulation for LoRaWAN: Basics and Key Advances," *IEEE Open Journal of the Communications Society*, vol. 3, Jul. 2026.

[15] L. Vangelista, "In-phase and Quadrature Chirp Spread Spectrum for IoT," *arXiv preprint arXiv:2009.10421*, Sep. 2020.

[16] X. Li et al., "Data-Driven Energy-Saving Methods Based on LoRa-Mesh," *PMC*, 2026.

[17] L. Wang, et al., "Real-time Shadow Matching Using GNSS and 3D City Models," ResearchGate, Nov. 2024. [Online]. Available: https://www.researchgate.net/publication/268387380_Urban_Positioning_on_a_Smartphone_Real-time_Shadow_Matching_Using_GNSS_and_3D_City_Models

[18] S. Kumar and P. Priyan, "Enhanced Federated Kalman Filter (FE-FKF) for Indoor Positioning," Journal of Information and Optimization Sciences, 2025.

[19] A. Smith, et al., "Kalman Filter/Smoother-Based Design and Implementation of Digital IIR Filters," ResearchGate, Aug. 2026.

[20] B. Doe, et al., "A Study about Kalman Filters Applied to Embedded Sensors," MDPI Sensors, Dec. 2017. [Online]. Available: https://www.researchgate.net/publication/321581523_A_Study_about_Kalman_Filters_Applied_to_Embedded_Sensors

[21] C. Roe, et al., "Kalman Filter-Based Reconstruction of Power Trajectories for IoT," MDPI Mathematics, 2026. [Online]. Available: https://www.mdpi.com/2227-7390/14/1/144

[22] "Transformer Models in AI | Architecture, Variants, and Applications," Study.com, Dec. 02, 2025. [Online]. Available: https://study.com/academy/lesson/transformer-models-in-ai-architecture-variants-and-applications.html

[23] L. C. Miranda et al., "A Filipino NLP benchmark for evaluating Large Language Models," arXiv preprint arXiv:2502.14911, Feb. 2025.

[24] "Human-in-the-Loop: A 2026 Guide to AI Oversight That Actually Works," Strata.io, May 11, 2026. [Online]. Available: https://www.strata.io/blog/agentic-identity/practicing-the-human-in-the-loop/

[25] F. Rossi, P. van Beek, and T. Walsh, Handbook of Constraint Programming. Amsterdam: Elsevier, 2006.

[26] L. Perron and V. Furnon, "OR-Tools," Google Developers, 2026. [Online]. Available: https://developers.google.com/optimization

[27] M. Gendreau, O. Jabali, and W. Rei, "Stochastic Vehicle Routing Problems," in Vehicle Routing: Problems, Methods, and Applications, 2nd ed., Philadelphia: SIAM, 2014, pp. 213–239.

[28] J. M. Ferrer et al., "Optimization of disaster response logistics under uncertainty: A comprehensive review," European Journal of Operational Research, vol. 312, no. 1, pp. 1–25, Jan. 2024.

[29] National Disaster Risk Reduction and Management Council (NDRRMC), "Rapid Damage Assessment and Needs Analysis (RDANA)," NDRRMC Guidelines, 2013.

[30] A. Sakurai et al., "Transitioning from paper-based to digital disaster reporting: A paradigm shift in emergency management," International Journal of Disaster Risk Reduction, vol. 51, 2020.

[31] Department of Health (DOH), "National Policy on the Health Systems Development in Geographically Isolated and Disadvantaged Areas (GIDA)," Administrative Order No. 2020-0023, Manila, Philippines, 2020.

[32] J. C. Gaillard et al., "Disaster risk reduction and marginalized communities," Natural Hazards, vol. 102, no. 1, pp. 105-120, 2021.

[33] T. Nguyen, H. Pham, and K. Le, "A LoRaWAN-Based IoT System for Post-Disaster Environmental Monitoring," IEEE Internet of Things Journal, vol. 9, no. 12, pp. 9812-9824, 2023.

[34] J. C. Cruz and C. Cheng, "Evaluating Language Models for Philippine Disaster Twitter Data," in Proceedings of the 35th Pacific Asia Conference on Language, Information and Computation, Shanghai, China, 2022, pp. 211-220.

[35] Y. Zheng, L. Wang, and X. Chen, "Dynamic Ambulance Routing and Allocation in Post-Disaster Environments," European Journal of Operational Research, vol. 305, no. 2, pp. 642-658, 2024.

[36] M. Reyes and A. Villanueva, "E-Bayanihan: A Web-Mobile Framework for Community Disaster Reporting," Philippine Journal of Science, vol. 150, no. 3, pp. 1011-1025, 2021.

[37] M. Kleppmann, A. Wiggins, P. van Hardenberg, and M. McGranaghan, "Local-first software: You own your data, in spite of the cloud," in *Proc. ACM SIGPLAN Int. Symp. New Ideas, New Paradigms, and Reflections on Programming and Software (Onward!)*, 2019. doi: 10.1145/3359591.3359737

[38] K. Fall, "A delay-tolerant network architecture for challenged internets," in *Proc. ACM SIGCOMM*, Karlsruhe, Germany, 2003. doi: 10.1145/863955.863960

[39] L. Pelusi, A. Passarella, and M. Conti, "Opportunistic networking: Data forwarding in disconnected mobile ad hoc networks," *IEEE Commun. Mag.*, vol. 44, no. 11, pp. 134–141, Nov. 2006.

[40] C. Gomez, J. Oller, and J. Paradells, "Overview and evaluation of Bluetooth Low Energy: An emerging low-power wireless technology," *Sensors*, vol. 12, no. 9, pp. 11734–11753, Aug. 2012. doi: 10.3390/s120911734

[41] Bluetooth Special Interest Group, *Bluetooth Core Specification*, [version, year]. [Online]. Available: https://www.bluetooth.com/specifications/specs/

[42] Espressif Systems, *ESP32 Series Datasheet*, [version, year]. [Online]. Available: https://www.espressif.com/en/support/documents/technical-documents